

ARCT · FY2025 Q4

Arcturus Therapeutics Holdings Inc. · 97 turns · 19 participants

CAST (IN ORDER OF APPEARANCE)**Neda Safarzadeh**

Vice President, Head of Investor Relations, Public Relations, and Marketing

Joe Payne

President and CEO

Dr. Alan Cohen

Chief Medical Officer

Lily Songo

Analyst, Learing Partners

Operator**Operator****Tom Schrader**

Analyst, BTIG

Whitney Ejem

Analyst, Canaccord

Yanan Zhu

Analyst, Wells Fargo

Yugal Nokomovitz

Analyst, Citi

Joe Payne

President and CEO

Adam Walsh

Analyst, Roth Capital

Evan Wang

Analyst, Guggenheim Securities

Miles Minter

Analyst, William Blair

Operator**Pete Stavropoulos**

Analyst, Cantor Fitzgerald

Unknown**Yale Jen**

Analyst, Laidlaw & Company

Yasmeen Rahimi

Analyst, Piper Sandler

OPERATOR

Thank you for your continued patience. Your meeting will begin shortly.

OPERATOR

If you need assistance at any time, please press star zero, and a member of our team will be happy to help you.

OPERATOR

Hello and welcome everyone joining today's Arcturus Therapeutics fourth quarter in fiscal year 2025 earnings call. At this time, all participants are in a listen-only mode. Later, you will have the opportunity to ask questions during the question and answer session. To register to ask a question at any time, please press star 1 on your telephone keypad. Please note this call is being recorded. We are standing by if you should need any assistance. And it is now my pleasure to turn the meeting over to Neda Safarzadeh, Vice President, Head of Investor Relations, Public Relations, and Marketing. Please go ahead.

NEDA SAFARZADEH Vice President, Head of Investor Relations, Public Relations, and Marketing

Thank you, Operator. Good afternoon and welcome to ArcTera Therapeutics Quarterly Financial Update and Pipeline Progress Call. Today's call will be led by Joe Payne, our President and CEO. and Dr. Alan Cohen, our Chief Medical Officer. Dr. Pat Shivakula, our CSO and COO, will join them for the Q&A session. Before we begin, I would like to remind everyone that the statements made during this call regarding matters that are not historical facts are forward-looking statements within the safe harbor provisions of the Private Securities Litigation Reform Act of 1995. Forward-

looking statements are not guarantees of performance. They involve known and unknown risks, uncertainties, and assumptions that may cause actual results, performance, and achievements to differ materially from those expressed or implied by the statement. Please see the forward-looking statement disclaimer on the company's press release issued earlier today, as well as the risk factor section in our most recent Form 10-K and in subsequent filings with the SEC. In addition, any forward-looking statements represent our views only as of the date such statements are made. Arcturus specifically disclaims any obligation to update such statements. And with that, I will now turn the call over to Joe.

JOE PAYNE President and CEO

Thank you, Netta. It's good to be with you again, everybody. I will begin today with an update on our ARCT 032 and ARCT 810 programs. These are the messenger RNA therapeutic candidates for cystic fibrosis and ornithine transcarbamylase deficiency, respectively. ARCT032 phase two trial is progressing with higher dose testing at 15 milligrams in four class one CF adults, which showed no safety concerns. The upcoming multi-month study will enroll patients in the US and internationally, and is designed to evaluate safety as well as look for early signs of clinical benefit, including improvements in lung function and quality of life. We are well on track to initiate dosing for this Phase 2 12-week study in the first half of this year and look forward to generating potentially meaningful clinical data for our CF program in 2026. ARC-T810 continues to advance toward pivotal development. We plan to study both adults with late-onset OTC deficiency and young children with the most severe forms. Type C regulatory meetings are scheduled during the first half of 2026 and are intended to provide clarity regarding our next steps in clinical development for our flagship rare liver disease program. I will now provide regulatory updates to our partnered COVID-19 vaccine program, also known as CoStave, where in January 2026, the UK Medicines and Healthcare Products Regulatory Agency, or MHRA, granted approval for CoStave, a self-amplifying mRNA COVID-19 vaccine for use in individuals aged 18 and older. Moving to ARCT 2304. This is our next generation star vaccine candidate for pandemic A slash H5N1 influenza. This program is contracted with and funded by BARDA. We completed a phase one study in 212 young adults and 80 older adults. Recent eight-month follow-up data after the initial vaccination showed that all three dose levels, 1.5, 5, and 12 micrograms, generated a durable immune response. The results also reinforced the SA mRNA platform's ability to drive meaningful cell-mediated immunity. Across all tested doses, the vaccine was well tolerated with no safety concerns reported. The data further validates our STAR SA mRNA platform. Also briefly, our lawsuit against AbbVie and Capstan Therapeutics filed on September 23rd, 2025, remains ongoing.

JOE PAYNE President and CEO

With that, I'll now pass the call to Alan.

DR. ALAN COHEN Chief Medical Officer

Thank you, Joe. It's certainly good to be here with all of you today. I'll begin with an update of our ARCT032 program. This is our messenger RNA therapeutic candidate for cystic fibrosis or CF pulmonary disease. ARCT032 utilizes Arcturus's Lunar Lipid-Mediated Aerosolized Platform, to deliver CFTR messenger RNA to the lungs. Expression of a functional copy of the CFTR mRNA in the lungs of people with CF has the potential to restore CFTR activity and mitigate the downstream effects responsible for the progressive lung disease and associated morbidity and mortality experienced by people with CF. We remain on track to initiate the dosing phase of our 12-week phase two clinical study in the first half of 2026. We recently completed one daily dosing of 15 milligrams of ARCT032 over 28 days in the third dosing cohort. This cohort included four CF adults with class one null mutations. And importantly, we observed no safety or tolerability concerns at this higher dose. Based on these encouraging data, the Independent Safety Review Committee permitted us to move forward with a phase two study that will evaluate ARCT032 over a 12-week period instead of just four weeks. We will include two functional pulmonary measures, spirometry, as measured by percent

predicted FEV1, as well as LCI, or lung clearance index, as measured by multiple breath washout. We will also include two quality of life measures, the CFQRR and the EQ5D5L, as well as serial high resolution CT testing to examine changes in airway wall thickness, air trapping, and mucus plugging scores. The 12-week study intends to enroll up to 20 participants with Class 1 CF mutations from clinical sites in the U.S. and abroad. The goal of this upcoming 12-week study is to establish a clearer picture of longer-term safety, 12 weeks versus four weeks, and begin to more comprehensively explore early signals of clinical efficacy, including how the CF participants feel and function. Now moving on to the ARCT 810 program. This is our messenger RNA therapeutic candidate for Ornithine Transcarbamylase or OTC deficiency. We look forward to aligning with regulators on our clinical development strategy for the OTC deficiency program. We are moving to broaden our development strategy to serve both adults and late onset disease and young children with the most severe forms of OTC deficiency. These are the patients who typically rely on liver transplantation for survival beyond early childhood. We continue active regulatory engagement to support pivotal trial designs across these two distinct populations. Our type C regulatory meetings with the health authorities, along with their feedback, remain on track for the first half of 2026, and we expect these interactions to help clarify our clinical development strategy for both adult and pediatric indications. Currently, we are very pleased with the momentum across both the CF and OTC deficiency programs and look forward to updating you as we progress throughout the year. I'll now pass the call back to Joe.

JOE PAYNE President and CEO

Thanks, Alan. We indeed look forward to initiating enrollment in our 12-week CF study and gain clarity and alignment with regulatory agencies for the next stages of development for ARCT 810. Now, with respect to Arcturus' financial position, We issued a press release earlier today, which includes financial statements for the fourth quarter and fiscal year ending December 31, 2025, and provided a summary and analysis of year-over-year performance. Please also refer to our most recent Form 10-K for more details on financial performance. Year-over-year, annual and quarterly revenue decreased \$70.3 million and \$15.6 million, respectively. These declines were driven by reductions in revenue from our CSL collaboration, reflecting lower supply agreement activity and a reduced number of development-based milestone achievements as CoastAid was commercialized. Annual and quarterly research and development expenses also decreased year-over-year by \$83.0 million and \$19.3 million, respectively, which was primarily driven by lower manufacturing and clinical costs, related to the Lunar COVID program, reflecting the program's transition from a development program to the commercial phase. Additional decreases relate to lower manufacturing for our Lunar CF and Lunar Flu programs, as well as lower clinical costs for our Lunar OTC program as we wrap up clinical activities. Clinical costs for phase two of our Lunar CF program partially offset these reductions as we remain focused on this clinical study. General and administrative expenses decreased annually, year over year, by \$6.7 million, due to reduced expenses for payroll and benefits, as well as share-based compensation. Quarterly general and administrative expenses increased year over year by \$1.6 million, due to an acceleration of employee stock options. Overall, we expect general and administrative expenses to continue to decrease during the next 12 months driven by lower share-based compensation expense. Cash, cash equivalents and restricted cash were \$232.8 million as of December 31, 2025 and \$293.9 million on December 31, 2024. Through disciplined execution and a strategic refocus on existing rare disease clinical programs in fiscal year 2025, Arcturus has extended our cash runway into the second quarter of 2028. In summary, the company remains in a strong financial position and has the cash runway needed to achieve multiple near-term value creating milestones for both therapeutic programs. With that, let's turn the time over to the operator for questions.

OPERATOR

Thank you. If you'd like to ask a question, press star 1 on your keypad. To leave the queue at any time, press star 2. Once again, that is star 1 to ask a question. Thank you. Our first question comes from Yasmeen Rahimi with Piper Sandler. Your line is open.

YASMEEN RAHIMI Analyst, Piper Sandler

Good afternoon, team. Thank you so much for the prepared remarks. Would love to understand, as you guys are going to be taking up the 12-week Phase II study, I think you guys spent a lot of time thinking about optimization for the study from dose selection to baseline measurements to patient selection. So maybe help us walk us through some of the optimizations that were included versus the initial study that was conducted in the fall that would be really helpful to go through. And then also, I know you work with the CF Foundation. Help us understand if you had a chance to work with them to warehouse a number of patients so as soon as you pick up the study, you could enroll quickly the patients for it. With that, I'll jump back into the queue.

JOE PAYNE President and CEO

Yeah, thanks, Yaz. Appreciate the questions, as always. With respect to how the 12-week study that we're initiating here soon is different from the four-week study, we have Alan on the line.

JOE PAYNE President and CEO

He'll highlight some of those differences so that the investors out there can understand.

DR. ALAN COHEN Chief Medical Officer

Sure, no problem. Thanks, Joe. Great question. So first things first, one of the main fundamental differences between The study that we've done with dose ranging from cohorts one through three is that cohort four is really going to be designed somewhat differently with the intent on it being more reproducible and stable, particularly as it relates to spirometric measures. We're setting parameters. We will not be enrolling a CF patient until they're stable with respect to baseline lung function measures, and this should help us allow us to observe true clinical signals by enhancing the noise-to-signal ratio. In addition, we are going to be including lung clearance index using MBW, which is not subject to variability issues that is sometimes seen in spirometry. And that's why LCI has really been used and preferred in children because of its ability to be reproducible and not patient or performance dependent. In addition, we're going to be looking over a 12-week period instead of just simply four weeks with the general thinking that the longer period of time will allow the drug that we're studying to manifest clinical benefits in the airway. Lastly, we are including not just LCI and using MBW and percent predicted FEV1 spirometry, but two additional quality of life measures, the CFQRR and the EQ5D5R, So two shots on goal for both pulmonary functions and quality of life, in addition to the HRCT studies reproducibly that we have used in the past and shown early signals of success. Hopefully that should be helpful to you.

JOE PAYNE President and CEO

Thank you, Andy. And we do continue to strengthen our relationship with the CF Foundation. They're keenly aware of the modulator non-responders in their community, both here in the U.S. and abroad. and so we work closely with them. In fact, the TDN themselves permitted our study to proceed after reviewing the first three cohorts of data into this fourth cohort, into this 12-week study, so they're closely involved and engaged.

UNKNOWN

Thank you.

OPERATOR

We'll move next to Pete Stavropoulos with Cantor Fitzgerald.

OPERATOR

Your line is open.

PETE STAVROPOULOS Analyst, Cantor Fitzgerald

Yeah. Hi, Joe, Alan, and team. Nice to see the progress. So, you know, you mentioned LCI for CF. Can you, you know, talk a little bit about this test? You know, sort of how sensitive is it? I think you mentioned it's not variable. Do you expect multiple readings at baseline? And importantly, you know, does it correlate with other endpoints you plan to use? And is it reliable at all stages of disease, say early versus late, or equally sensitive? And if so, you know, how would that impact your enrollment criteria for the phase two?

JOE PAYNE President and CEO

Oh, it's a great question, Pete. LCI is definitely another lung function measurement that we're, data that we're collecting in this 12-week study. It's definitely more sensitive. Alan, perhaps you can discuss about the multiple readings and how well it correlates and how reliable it is.

DR. ALAN COHEN Chief Medical Officer

Sure, of course. And a great question. Thank you for asking about this. LCI, as I mentioned, doesn't have the subject variability issues of spirometry, although as we all know, spirometry has been a more traditional endpoint measure for most pulmonary drugs that have been approved both in the United States and abroad. And that's really why LCI historically in CF has been used in children who can't always perform spirometry reproducibly. It's a much more passive maneuver requiring just normal tidal breathing, so comfortable breathing in and out. And as a result, reproducible measure for, more importantly, the most early and more subtle changes in the smallest airways. So why we're focusing on LCI in addition to spirometry is not just its ability to be reproducible, but it also is measuring another component of the airway that we believe we can have more discernible benefit for earlier and in a smaller number of subjects. Spirometry measures can change slightly, as well in these larger airways, and these are more central airways, but it does require more active engagement to be sufficiently reproducible. That's why I was alluding earlier to the controls that we're putting into this study as to making sure that a patient has reproducibility in screening and baseline before going into drug dosing so that we have a more accurate reproducible baseline with which to compare subsequent changes over the 12-week course of the study.

JOE PAYNE President and CEO

All right.

PETE STAVROPOULOS Analyst, Cantor Fitzgerald

Thank you for that. And also just a question on the OTC program. You know, as you're having your discussions with regulatory authorities, you know, about a registrational trial, how has their reception been for certain biomarkers and assays, you know, like ureogenesis function using the nitrogen 15 isotope or any other biomarker they may be less familiar with? And, you know, what's sort of the base case outcome for a study design?

JOE PAYNE President and CEO

Well, those are all great questions, and that's currently where we are. We're actively engaged in preparing for these Type C meetings in the next few months. We're going to be in a much better position to answer those with granularity and specificity. But that's exactly what we're currently engaged with the FDA on, is the future study approaches, the design, the size, the scope of the study, understanding what they would like to see before we can proceed further

into the clinical development. We'll have that regulatory clarity shortly. That's our intent. That's our aim. They're all good questions, but we'll be in a better position to answer those in a few months.

PETE STAVROPOULOS Analyst, Cantor Fitzgerald

All right. Thank you very much, and congrats on the quarter and the progress. Thanks, Pete.

OPERATOR

We'll move next to Seamus Fernandez with Guggenheim Securities. Your line is open.

EVAN WANG Analyst, Guggenheim Securities

Hi, guys. This is Evan Wang on for Seamus Fernandez. Just a few from us. Looking forward to the start of the 12-week CF study. Just wanted to clarify, so it sounds like dosing is proceeding at the 15-mg dose cohort. Just curious, what drove the decision to proceed with the 15-mg dose over 10? Was there any evidence of dose response on any of the lung function measures or CT scans? Second, In terms of planned enrollment for CF, you mentioned both U.S. and international recruitment. Just curious how you're thinking about maybe the split between U.S. and abroad.

JOE PAYNE President and CEO

Thanks. A couple good questions. First of all, a point of clarification, just so it's not misunderstood. We have achieved safety and tolerability data for the 15 milligram level, but we're actually initiating our fourth cohort at the 10 milligram level. We were fortunate to see early efficacy, some early clinical signals in our smaller cohorts at 10 milligrams. So we're going to continue that and see if we can see improvements as we extend the study in larger numbers. With respect to the plan to enroll, in the U.S., and we are enrolling internationally, including Europe and the Middle East, but Alan can maybe share some of the strategies as to why we're expanding our enrollment in other countries.

DR. ALAN COHEN Chief Medical Officer

Sure. Thank you, Joe, and a great question. So the plan right now, we've actually expanded our U.S. sites, so we have additional U.S. sites ready to go for the cohort 4 study, which we will be initiating shortly, and have added, as Joe mentioned, European as well as Middle Eastern sites. The intent there is to really go into parts of the world where there is a higher preponderance of null or type 1 CF genetic mutants so that there is a larger percentage within those populations, and it should yield a greater likelihood of identifying engaging and enrolling patients in a much more timely manner over the course of the calendar year.

EVAN WANG Analyst, Guggenheim Securities

Just one follow-up from me, then. Just between the 15 and 10-mg dose, because you mentioned improvements at the 10-mg, anything you can share in terms of the responder rate at the 15-mg relative to the 10? Did you see a response on FEV? I guess in terms of similar benchmark, you presented the earlier data.

JOE PAYNE President and CEO

Thanks. No, yes. We've completed the dosing phase for the third cohort at 15 milligrams, and we've collected sufficient safety and tolerability data to share with the Safety Review Committee to allow us to permit us to proceed into the 12-week study. With respect to the other data, that data collection process is ongoing. and that's where we are. But what was significant was to allow us to proceed into the fourth cohort.

UNKNOWN

Got it. Thank you. Thanks, Evan.

OPERATOR

We'll move next to Lily Songo with Learning Partners.

LILY SONGO Analyst, Learning Partners

Your line is open. Hi. Good afternoon. Thank you for taking my question. So two questions for me. The first one on the CCF program. Thinking about, again, moving on to the 12-week study, what was the rationale to stop at 15 mg given that you haven't seen any safety signal? And would there be enough design flexibility to potentially increase the dosing as you accumulate data? And then I have a follow-up question on the OTC program.

JOE PAYNE President and CEO

Yes, we definitely have the flexibility to increase dosing, and we're very happy that we already have the ability to increase to 15 mg if necessary. But just to maybe reiterate that 10 milligrams showed early signs of working in the second cohort, so we're simply extending the duration of that treatment in a larger cohort. And we're now pleased that we have the flexibility to increase dosing to 15 milligrams if needed. It's unlikely that we will need to increase dosing even further, but we would have flexibility to do so if needed. We did dose up to 27 milligrams in our early trials in human volunteers. So there is some headroom there to expand further, but we feel confident in the dose that we've selected at 10 to start this fourth cohort.

LILY SONGO Analyst, Learning Partners

Great. Thank you. Maybe just a follow-up. For the OTC program, Do you expect a bifurcated regulatory path for both the pediatric and then the more adult patient population?

JOE PAYNE President and CEO

Yeah, I think that's safe to assume that the younger children will be treated differently than the older stable adult population simply because the need is far more severe in younger children. It's a different population. But anything to add to that, Alan?

DR. ALAN COHEN Chief Medical Officer

Yeah, I mean, in many ways, I think where we are with the program, it's important to note that since there are survivors with an OTC deficiency into adulthood, even though it's vastly different in terms of the medical complexity and the ability to be functional without needing a liver transplant, we were able to show up to this point safety, tolerability, and we believe early signs and signals of efficacy. Now, in our discussions with the FDA on looking at a younger population where survival tends to be quite bleak beyond preschool and school age, the intent there is to really see if we can leverage the most recent guidances that the FDA has shared about their willingness to look into ultra-rare populations with severe outcomes and their willingness to be more, give more latitude and flexibility in terms of how one conducts those studies, as long as there's sufficient safety and tolerability, and obviously efficacy. So, we're encouraged by the early conversations that we're having with the agency right now, and we'll continue to pursue those to the benefit of the OTC deficiency community.

OPERATOR

Great. Thank you for the added color.

UNKNOWN

Thanks, Lily.

OPERATOR

We'll move next to Yanan Zhu with Wells Fargo. Your line is open.

YANAN ZHU Analyst, Wells Fargo

Thanks for taking our questions. Maybe first on the CF program, I think you mentioned one effort in Core 4 is to obtain a stable baseline. Remind us why that's important. How are you going to go about and do that? And also curious, during the three-month trial period, could it be possible that patient had a stable baseline, but for some reason their reading could begin to fluctuate or drift after having had that stable baseline within a three-month period? Thank you.

JOE PAYNE President and CEO

Thanks, Yanan. Yeah, cohort four is definitely different with respect to the the first three cohorts, and that we've designed it with a stronger baseline. Alan, maybe you can comment further there.

DR. ALAN COHEN Chief Medical Officer

Yeah, so what we've done here, and it's a great question. You know, as best as we can, one should realize that because of the complexity of the lung disease and the impact that it has in the various segments of the lung at any given time, What we're looking for at baseline with these adult patients with CF is at least enough reproducibility and as small a possible variability within the period of time that we screen to baseline to get them enrolled that we believe that we have as accurate a representation of where they are at that moment in time as far as their lung disease is advancing and progressing. It's not unexpected in the course of a three- or six-month period that Many of these patients may have an acute pulmonary exacerbation or an acute illness. And we're prepared to handle that. But what we wanted to try to mitigate and deal with was the variability that is seen not uncommonly in patients with COPD, cystic fibrosis, asthma, et cetera, particularly those who are vulnerable and chronically infected with pathogens. So we're really tightening up the enrollment criteria. We're setting parameters. And that's why we believe we're just looking to make sure that it's reproducible and stable at that baseline. And we're not going to be enrolling CF patients that may be in the early throes of a pulmonary exacerbation or an acute illness. And sometimes just even modest changes can be an early signal for that. We want to enhance the noise to signal ratio. And once again, one more thing to think about, unlike lung clearance index, Spirometry really does require a consistent study subject performance. It's not a passive maneuver. So anything that we can do to make sure that a patient coming into our study has a representative baseline is really what we're trying to achieve here.

YANAN ZHU Analyst, Wells Fargo

Great. That's super helpful. Two more questions, if I may. On the OTC deficiency regulatory interaction front, Could you lay out for us, to you and to the principal investigators, what would be considered an exciting outcome with regard to the alignment and what perhaps also what might be the base case scenario for the alignment? And lastly, I wanted to, wondering if you have any update on the COVID-19 vaccine initiatives between you and CSCL. Thank you.

JOE PAYNE President and CEO

Yeah, I can take those questions, Yanan. So we're definitely working with the FDA to establish clarity, right? That's the main objective. So as investors are looking at what are we trying to achieve with these Type C meetings, it's clarity. We want a clear path forward to helping both the the children in need with high unmet medical need, more severe disease, and the stable adults, right? So we're presenting our case, we're presenting our data, and basically looking to have clarity in the path forward. Anything to add there before I comment on CSL, Alan?

DR. ALAN COHEN Chief Medical Officer

No, and, you know, obviously in these patients, the goals and objectives for the children are quite different because of the severity in nature and labile nature of the disease in children as opposed to the much more stable adults. Even with current standards of care, the goals and objectives of a pediatric program would be to try to mitigate as best we can the damaging neurological and developmental issues that they contend and deal with on a daily basis just by having OCT deficiency. Despite the fact that there are now therapies like ammonia binding agents and abilities to control diet. So success for us would be, as Joe said, a clear path forward with agreed endpoints and goals and objectives that are not only attainable, but are potable and acceptable to the families and patients, as well as the caregivers and their current standards of care.

JOE PAYNE President and CEO

And the details of that, we'll be able to provide that granularity in a couple months. But I appreciate the question. With respect to your CSL-related question in CoastAid, yes, we definitely made some progress for the CoastAid asset and for the platform in the United Kingdom. It was good to see that we advanced CoastAid to the point of approval and licensure in the United Kingdom. But the present administration here domestically has made it challenging to progress CoastAid to licensure or approval in the near term here in the US. So because of this, and understandably, CSL and Arcturus are in active discussions regarding our collaboration, and we're going to be able to provide more color on that in the coming months.

UNKNOWN

Great.

JOE PAYNE President and CEO

Thanks for all the color.

OPERATOR

Thanks.

OPERATOR

We'll take our next question from Miles Minter with William Blair. Your line is open.

MILES MINTER Analyst, William Blair

Hey, thanks, guys, for the questions. First one's just on the fact that I think you're deciding between the 10-meg and the 15-meg dose for the 032 Phase 2 trial. I think you're waiting on 15-meg efficacy data. You've got the safety and tolerability, obviously. I'm trying to understand what... puts and takes off for selecting dose moving forward. Like if you did get this efficacy data in for the 15 megs and it looked fine, but you have more data in 10 megs, like which dose or would you take both doses forward? I'm trying to understand your logic between. Sure, sure.

JOE PAYNE President and CEO

So it's relatively straightforward in that 10 milligrams showed early signs of working in the second cohort. So we're simply extending the duration of treatment into a larger cohort and extending that duration. We now have the flexibility to increase the dose. That's what this means. And only if needed. There's many that are suggesting that efficacy is achieved through simple duration by extending the treatment, and we're going to prove that. And at some point, because it's an open-label study, if we feel we need to increase the dose, we can't. I also remind folks that it's a fairly pricey product to manufacture, so if we can keep cost of goods down, that's also a good benefit by keeping the dose down at around 10 milligrams as well. But it's simply based on data that at 10 milligrams we saw progress,

so we think we're going to extend the duration and the size of the study to see if it works in that context, in the fourth cohort. And we now have the flexibility to increase the dose if needed. Cool.

UNKNOWN

Thanks. Thanks, Miles.

OPERATOR

We'll take our next question from Adam Walsh with Roth Capital. Your line is open.

ADAM WALSH Analyst, Roth Capital

Hi, good afternoon. Thanks for taking my questions. The first on 032, the CF product. You've talked about the first week in your clinical trials or two of being an onboarding phase where the drug is sort of getting through the congested airways. So in a 12-week study, you'd have roughly 10 productive weeks versus maybe two or three in the 28-day study. How should we think about that from an extra exposure time as it relates to FEV1 and mucus clearance versus what we maybe saw at 28 days?

JOE PAYNE President and CEO

Thank you. Adam, that's a thoughtful question, and you're absolutely right. We view that in our 28-day study, we observed consistently that there was a one- to two-week onboarding phase. And so in reality, there's really only a two-week healing phase for that 28-day study. So as we go to the 12-weeks study, assuming a similar onboarding rate of one to two weeks, then you have 10 full weeks of potential healing and efficacy reads. So it's a difference between 2 and 10 weeks. So under that, from that perspective, it is a significant improvement. But if you look at the study as a whole, we're going from 4 weeks to 12 weeks. It's a threefold difference. That's also a significant bump. But your observation is sound, and we agree with you that it's not only threefolding the time of the study, but increasing the duration of potential healing and efficacy with this 12-week study.

ADAM WALSH Analyst, Roth Capital

Thanks. And then, if I could, one on 810 for OTC deficiency. Phase 2 patients stayed on standard of care. So, to my knowledge, we haven't seen whether patients can actually reduce scavengers or loosen dietary restrictions on 810. Is that something you plan to build into the pivotal and how important you think that kind of functional clinical data will be for the approval conversations versus the biomarker package alone?

JOE PAYNE President and CEO

Thank you. That exact question is in the agenda of our Type C meetings this year. So for both the children early onset and also with the stable adult population. So it's the right question. We'll be able to provide the answer to that question in more detail in a couple months after our Type C meetings are completed and we've received their feedback.

OPERATOR

Thank you.

UNKNOWN

Thanks, Adam.

OPERATOR

We'll take our next question from Whitney Ejem with Canaccord. Your line is open.

WHITNEY EJEM Analyst, Canaccord

Hi, guys. Thank you for taking your questions. This is Angela on for Whitney. Maybe just to follow up on the program again. So, understand you're progressing with the 10 milligrams for the 12-week study, but are you still looking at efficacy in the 15 milligrams? I guess, like, what efficacy are you looking at, and will you be sharing that data at some point, either in a PR or at a medical conference? And the second part is, would it be possible that you would run another 12-week study using the 15 milligrams? When you talk about the optionality to dose up, was that part of the current study?

JOE PAYNE President and CEO

Yeah, it's unlikely that we'll be sharing any more data from the first three cohorts. We are focused on the fourth cohort. This is almost like a phase 2B type study where we believe that our best shot on goal here is in a 12-week duration. and of 20 study, and that's where we're focusing on. So I think that's where the investors should also be focused on. But you've asked, are we collecting data for 15 milligrams? Absolutely. Is there a time and a place to share that data? Yeah, very likely.

JOE PAYNE President and CEO

But we remain focused on the 12-week study is the short answer to your question.

WHITNEY EJEM Analyst, Canaccord

Got it. And then will you potentially be dosing up to 15 milligrams in the 12-week study, or would that be a second study?

JOE PAYNE President and CEO

Only if needed. It's an open-label study. We have the flexibility to do it based upon what's happened in the third cohort, which is great. And we still haven't collected all the data, of course, and that additional data could sway our minds either direction. But right now we're initiating the fourth cohort at the 10-milligram dose level, and we have the flexibility now to increase it if needed.

UNKNOWN

Thank you.

JOE PAYNE President and CEO

Thank you, Angela.

OPERATOR

And we'll take our next question from Yugal Nokomovitz with Citi. Your line is open.

YUGAL NOKOMOVITZ Analyst, Citi

Hi, guys. This is Juwan Kim on for Yugal. Thanks so much for taking our question. I kind of have a multi-part question, but I believe in the Phase 1b, you had also assessed LCI. But can you remind us what you had seen previously on the LCI, and how should we be thinking about the level of improvement expected in the Phase 2b? And as you were headed into the dose-ranging Phase 2, did the FDA get a chance to look at the LCI data by any chance?

JOE PAYNE President and CEO

Yeah, you're correct that we did look at LCI in the early Phase 1 studies, and those Phase 1 was in healthy adults, Phase 1b was in CF participants and we established safety and tolerability that was sufficient to advance it. So LCI

was only after a single administration or two administrations in Phase 1B. I'll defer to Alan in terms of that data from Phase 1B. With respect to LCI as an endpoint, I think I'll also defer to Alan. He can talk about what the literature shows or the magnitudes of changes in LCI to get approval previously in children, et cetera. Go ahead, Alan.

DR. ALAN COHEN Chief Medical Officer

Sure. No problem at all. And it's a great question. I wasn't around historically when the early experience with our study drug in those first few cohorts going from healthy to CF adults were ongoing, but I have had a chance to review the data. You know, the challenge of it is, once again, it was small numbers of patients. In the way we're constructing things now, the plan is to benefit from that prior experience and perhaps do it in a more refined way. Right now, we also have the support of the CF Foundation, who is doing a natural history study. So being able to align with the CF Foundation and the Therapeutic Development Network on their prospective study that they're doing right now in patients with null mutations and those who are unable to tolerate current modulators allows us to align a study in a way so that when that natural history study becomes available in the coming months and year, that we can have access to that as a comparator group, which was something that was not available just a few years ago. The other piece about this I think is that there's increasingly a recognition of not only the sensitivity and the value of LCI measures in adults, but an appreciation for what may be the range and magnitude of change one needs to see in order for there to be an appreciation for a significant, meaningful change over time. So right now, I think we're doing the study in a manner that should optimize its utility and value. And we're also doing it in an alignment and way that should be commensurate with current standards and practices, which was certainly done before, but just not done as robustly. and as long-term. And really, I think in this case, just like with percent predicted FEV1 and spirometry, we will be benefiting from a cohort of upwards to 20 patients and an examination of data, not just over a four-week period, but a 12-week period. So both of those elements, in addition to having more normative data to compare to in this exact population, should make interpretation and value of this data much greater than it was previously appreciated.

UNKNOWN

Hopefully that's helpful to you. Yes, thank you very much. Appreciate it.

OPERATOR

We'll move next to Tom Schrader with BTIG.

OPERATOR

Your line is open.

TOM SCHRADER Analyst, BTIG

Good afternoon. Thanks for all the updates. There have been a lot of questions. I assume you've been looking at a lot of pictures of mucus plugs with your data. Do you have a sense of what fraction of patients show a measurable difference? Is it very all over the place, all over the lung. And do you have any patients where it's possible to calculate that in 12 weeks the mucus plug should be gone? Did you have that level of data? And I have a quick vaccine follow up.

JOE PAYNE President and CEO

Well, we saw four out of six subjects in the second cohort. That was a 10 milligram cohort that showed a reduction of mucus plugs after just 28 days. So that was definitely encouraging. there was one subject that attributed the increase in mucus plugs to humidity and the sensitivity to humidity. And whether that's an outlier or whether that's simply a non-responder, we will learn as we collect more data, especially in this extended study that's larger. But it seems that a majority of these subjects respond differently. with a reduction in mucus plugs, and we feel that that's important to

mention, and that's an early sign of, you know, some clinical activity. With respect to your other question, maybe you could restate it, and we can turn that over now.

TOM SCHRADER Analyst, BTIG

Just a sense for, you would have had some sense of rates of clearance. Do you have confidence that 12 weeks is enough, or is that more just the next trial you're allowed to do?

JOE PAYNE President and CEO

That's the big question. Clearly, we saw something in 28 days, so there's reason to be confident that we'll see it in an extended study, larger study. And it's not just a larger extended study, but we're looking at a better study, a study that's designed more tightly with respect to numbers and baseline steadiness. But anything to add, Alan?

DR. ALAN COHEN Chief Medical Officer

Yeah. Yeah, so I was going to say, you know, the field of using high-resolution CT scanning as a complementary way of assessing progression of disease has been used for some time. I think the technologies, particularly artificial intelligence and an ability to standardize the reads now is vastly different and in many ways much better than it was as recently as just a few years ago. So I think the field is emerging. I would be... I just want to make sure that it's understood that, you know, the time constants and the damage that occurs in the lungs of people with cystic fibrosis is really quite patchy. And what's remarkable, I'm a former lung transplant physician from Washington University, and I'm one of the few people that has had the chance to actually examine the lungs of people who underwent a lung transplant and sustained significant bronchiectatic damage to the point of needing a lung transplant to survive. What's impressive about the disease is the very patchy nature, both in the upper and lower lobes, and you can have a completely bronchiectatic destroyed segment of lung right next to a completely normal appearing functional segment of a lung. So I believe that a drug that's administered by inhalation over long periods of time will eventually have the ability in large part to reach many, if not all, of the segments that we need to, but I have no delusion as to whether or not there will be a consistent, almost a vacuuming of the lungs and an ability to remove and diminish mucus plugging. I think the time constants, the ventilation perfusion matching, and just the nature of what's going on actively day to day with infections will really always make this challenging. Even mucolytic drugs like Pulmozyme, even in patients who are using those drugs one or more times a day, will still be found to have significant mucus plugging in certain segments of the lung at any given time, and it will change from period to period from x-ray to x-ray. So, the goal here is, I think, the long term. It's the totality of what we're able to observe, and the plan is to be able to correlate those improvements with things like lung clearance index, which clearly has, and someone was alluding to this in their question earlier, That's one of the measures that we believe since we've already shown improvements in high-resolution CT scan mucus plug burden, it wouldn't be surprising to see and observe changes in lung clearance index commensurate with that since those two measures actually go hand in hand. So hopefully that's helpful to you.

TOM SCHRADER Analyst, BTIG

No, that's great. Very useful. And then just a quick follow-up on your pandemic flu vaccines. You haven't said a lot, but is your safety – just qualitatively in line with a protein vaccine, or does it look more like an mRNA vaccine? Do you have any sense of how things are going to look?

JOE PAYNE President and CEO

Yeah, very similar. And with respect to safety and tolerability, the safety and tolerability profile was similar to other vaccine technologies. We are a lower dose technology to conventional mRNA vaccines. So what we've seen consist-

ently is any dose-related toxicologies will be beneficial to the self-amplifying mRNA tech from Arcturus. But in general, it's similar to other vaccine technologies from a safety and tolerance.

JOE PAYNE President and CEO

Okay, thanks.

OPERATOR

Yeah. We'll move next to Yale Jen with Laidlaw & Company. Your line is open.

YALE JEN Analyst, Laidlaw & Company

Good afternoon, and thanks for taking the question. In terms of the CF next 12-week study, and you're looking for a more stable baseline, would that increase the time and the size for the screening for the study? And also, how long duration to define as a more stable baseline before the patient will be eligible for the study? And then I have a follow-up.

JOE PAYNE President and CEO

Yeah, we're definitely not enrolling a patient until they are stable with respect to the baseline lung function measures. The duration of that stability time, do you want to comment on that, Alan?

DR. ALAN COHEN Chief Medical Officer

Yeah, I mean, you know, and this is a great question, and it's not a simple answer, so let me see if I can give as appropriate an answer as possible. The challenge with CF patients is that there's actively, chronically something going on at any given day, and I think the biggest challenge in trying to identify and enroll patients into a study like ours where, you know, you're looking for modest improvements over, in the case of our study's first three cohorts, over a four-week period to now a much larger group of patients over now a 12-week period, we do want to make sure that we're not just identifying the right patients, but we're also catching them at a time in the course of their disease where they're at least stable enough so that we can introduce our drug and begin the baseline assessment so that when we get to the end, we at least have a comparator that's stable, reasonable, and discernible. So right now, it's possible based on the way we've constructed this study, and I've done this in other studies as well, where we may delay the enrollment of a patient by a few weeks or a month just simply because there seems to be something acute going on, whether they're getting acutely ill and we just happen to be catching it during the course of the baseline and the screening, or they may be in the late stages of resolving their illness and we're seeing enough of a difference between two measures over a relatively short period of time, that it may just speak to the unstable nature of what's going on. We really just are looking to get the best patients at the best period of time so that we have an equivalent baseline with which to work from. And that's really the intention. And it may mean a few weeks or a month longer to enroll a given patient, but it shouldn't add much in terms of the time. I think the goal here is to really find the best patient at the best time. Hopefully that helps you.

YALE JEN Analyst, Laidlaw & Company

Absolutely. That's very helpful. Maybe just a quick one on the OTC. I know it's still early a little bit, but was the company intended, at least the desire to potentially starting both the adult and pediatric trial relatively concurrently, or there's any prior to one before the other?

JOE PAYNE President and CEO

I think the intent is always to get into the children with high-end medical need as fast as possible, but we're going to adhere to the advice and feedback from the FDA from these Type C meetings. So we'll be able to answer that with more granularity in a few months.

YALE JEN Analyst, Laidlaw & Company

Okay, maybe the last question. Does CSL intend to launch... cost of it in UK or do we know that? And thanks.

JOE PAYNE President and CEO

I would refer to everyone on the call to see what they've said publicly at their recent investor calls and their filings. We don't anticipate anything soon out of the United Kingdom commercially from CoastAid, but I refer you to their statements that they're saying publicly. We'll see where we are with the CSL agreement and our discussions with them in the coming months.

YALE JEN Analyst, Laidlaw & Company

Okay, great. Thanks a lot and congrats on the progress.

JOE PAYNE President and CEO

Thanks, Yale.

OPERATOR

And this does conclude the question and answer portion of today's program. I would now like to hand the call back to Joe Payne for any additional or closing remarks.

JOE PAYNE President and CEO

Thank you, operator. I'd just like to thank everyone for attending today's call. If we see you in the I think conference season's ahead. We look forward to catching up with you there. Have a good afternoon, everybody.

OPERATOR

Thank you. This brings us to the end of today's meeting. We appreciate your time and participation. You may now disconnect.